

Software Project Management (SPM) (CSC415)

Course Instructor
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Chapter 6: Monitoring and Control

Introduction

- In a project life cycle, monitoring and controlling is the process of regularly observing and tracking the progress of your project and making any necessary proactive corrections.
- This stage is vital in ensuring your project stays on track and within budget.
- In fact, a staggering 70% of projects fail to deliver their intended outcome.
- But when a proper management process is put in place, this number drops drastically to a failure rate of 20% or below.

Project Monitoring

- The project monitoring definition is pretty simple: It's the process of regularly observing and tracking the progress of your project.
- This is the first step in the project monitoring and controlling phase and involves collecting data and information about various aspects of the project.
- Plus, it's essential if you want to ensure project success and hit your objectives.

Project Monitoring

- Once work schedules have been published and the project is under way, attention must be focused on ensuring progress
- This requires ***monitoring***
 - *What is happening?*
 - *Comparison of actual achievement against the schedule*
 - *Revision of plans and schedules to bring project as far as possible back on target.*

Project Monitoring

- Project monitoring activities can include things like:
 - *Tracking project milestones and deliverables*
 - *Checking the project's performance is on track to meet goals, objectives, and Key Performance Index (KPI), and developing performance metric reports*
 - *Monitoring project manager performance metrics, so nothing falls through the cracks*
 - *Checking the project schedule and timeline is on track*
 - *Assessing the project budget and costs compared with the forecasts*

Project Monitoring

- *Staying on top of the project scope and making sure scope creep doesn't happen*
- *Carrying out an overall quality control assessment, and conducting quality reviews (and creating reports)*
- *Watching out for any general issues that arise, and building an issue log*
- *Conducting risk assessments and producing risk management plans*
- *Setting up progress meetings and conducting status reports and reviews*

Project Monitoring

- **Project Monitoring focuses on:**
 - *Measuring planned performance vs. actual performance.*
 - *Identify ongoing projects performance and prevent if any problems exist.*
 - *Keeps timely accurate information based on project progress.*
 - *Forecasts update on current costs and project schedule.*
- **There are some question to be answered as:**
 - *Why do we monitor?*
 - *What do we monitor?*
 - *When to we monitor?*
 - *Where to monitor?*
 - *How do we monitor?*

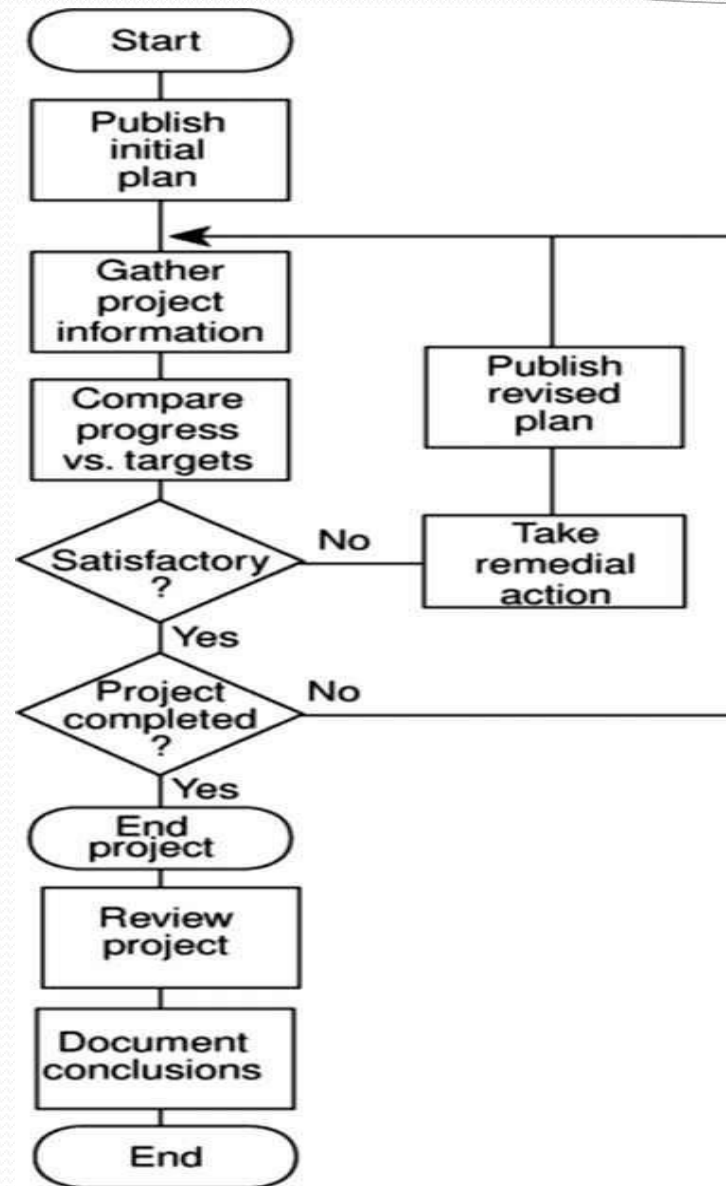
Project Monitoring

- ***Why do we monitor?***
 - To detect and react appropriately to deviations and changes to plans
- ***What do we monitor?***
 - Men (human resources), Machines ,Time, Materials, Tasks, Money, Quality/Technical Performance
- ***When to we monitor?***
 - End of the project, Continuously, Regularly, Logically, While there is still time to react, As soon as possible, At task completion, At pre-planned decision points (milestones)
- ***Where do we monitor?***
 - At head office?, At the site office?, On the spot?, Depends on situation

Project Monitoring

- *How do you monitor?*

- Through meetings with clients, parties involved in project (Contractor, supplier, etc.)
- For schedule - Update CPA, PERT Charts, Update Gantt Charts
- Using Earned Value Analysis
- Calculate Critical Ratios
- Milestones
- Reports
- Tests and inspections



Suggested Framework for Project Monitoring

Project Monitoring

- The project monitoring process enables project managers and stakeholders to make sure everything is ticking along as planned.
- It also allows you to identify any issues, risks, and deviations from the project plan and then address them before they get out of hand
- There are two ways for tracking the performance:
 - *Setting Check Points*
 - *Collecting Data*

Setting Check Points

- **Based on regular time intervals**
 - *Can be weekly or monthly or quarterly.*
 - *Depend on what to check and how to check.*
- **Based on a particular event**
 - *At the end of each activity.*
 - *In the middle of a critical activity.*
- **Should be set before the plan was published**
 - *Make sure everyone knows when and what the check points are.*

Collecting Data

- As a rule, the managers will try to breakdown activities into more controllable task of one or two weeks duration.
- But it is still necessary to gather information about partially completed activities and in particular forecasts how much work is left to be completed.
- Where there is a series of product, partial completion of activities is easier to estimate by counting the no. of records specification or screen layouts produced.
- Tracking can be done through:
 - *Partial Completion Report*
 - *Risk Report*

Partial Completion Report

- Indicate the work done by the personnel and the time spent on the work.
- Generally weekly time sheets are used to represent partial completion reports as they provide good information about what has been achieved.
- Optional information from Partial Completion Report:
 - *Likelihood of failing to complete the task by the scheduled date*
 - *Estimated time of completion*

Risk Report

- One popular way of overcoming the objections to partial completion reporting is to avoid asking for estimated completion dates.
- But to ask instead for team members' estimate of the likelihood of meeting the planned target date.
- Indicate the likelihood of meeting the scheduled target date
 - *Instead of asking the estimated completion date*

Risk Report

- Use the traffic-light method
 - *Identify the key (First Level) elements for assessment in a piece of work;*
 - *Break these key elements into constituent elements (second level)*
 - *Assess each of the second level elements on the scale green for “on target” amber for “not on target but recoverable” and red for “not on target and recoverable only with difficulty”*
 - *Review all the second level assessments to arrive at first level assessment*
 - *Review first and second level assessments to produce an overall assessment*

Risk Report

Activity Assessment Sheet

Staff : Zobel

Ref:IoE/P/100

Activity: Code and test module A

Week number	13	14	15	16	
Activity summary	G	A	R		
Component					Comments
Screen handling procedures	G	G	A		
File updating	G	A	R		
Compilation	G	G	A		
Run test data	G	A	A		

Visualizing Progress

- Having collected data about project progress, a manager needs a way for presenting that data to greatest effect.
- Presenting the collected data in a way that is easy to understand which helps to easily identify the problem activities or areas that need to be taken care of.
- Progress can be visualized through:
 - *Gantt Chart*
 - *Slip Chart*
 - *Ball Chart*

Gantt Chart

- It is one of the oldest and simplest technique for tracking the project visually.
- It is a static picture showing the current progress of activity indicating scheduled activity date.
- An activity bar chart showing
 - the activities, their scheduled dates and duration
 - the reported progress of the activities;
 - 'today cursor' i.e. today's task

Gantt Chart_Example

Website Development Process

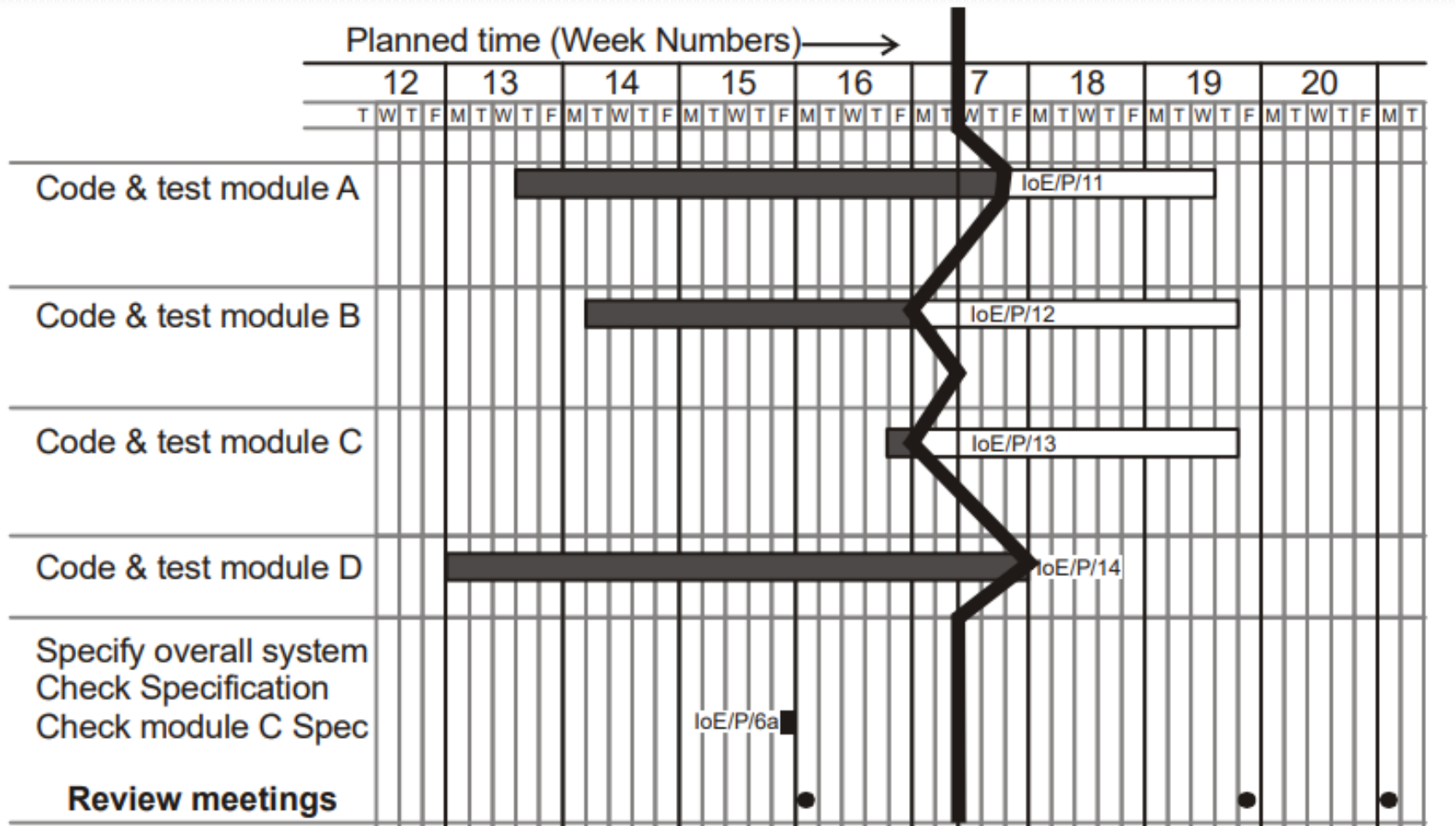
Gantt Chart

[illegible]

Slip Chart

- Provides a more striking visual indication of those activities that are not progressing to schedule.
- The slip line indicates those activities that are either ahead or behind the schedule
- Too much bending indicates a need for rescheduling of the overall plan
- The more the slip line bends, the greater the variation from the plan.

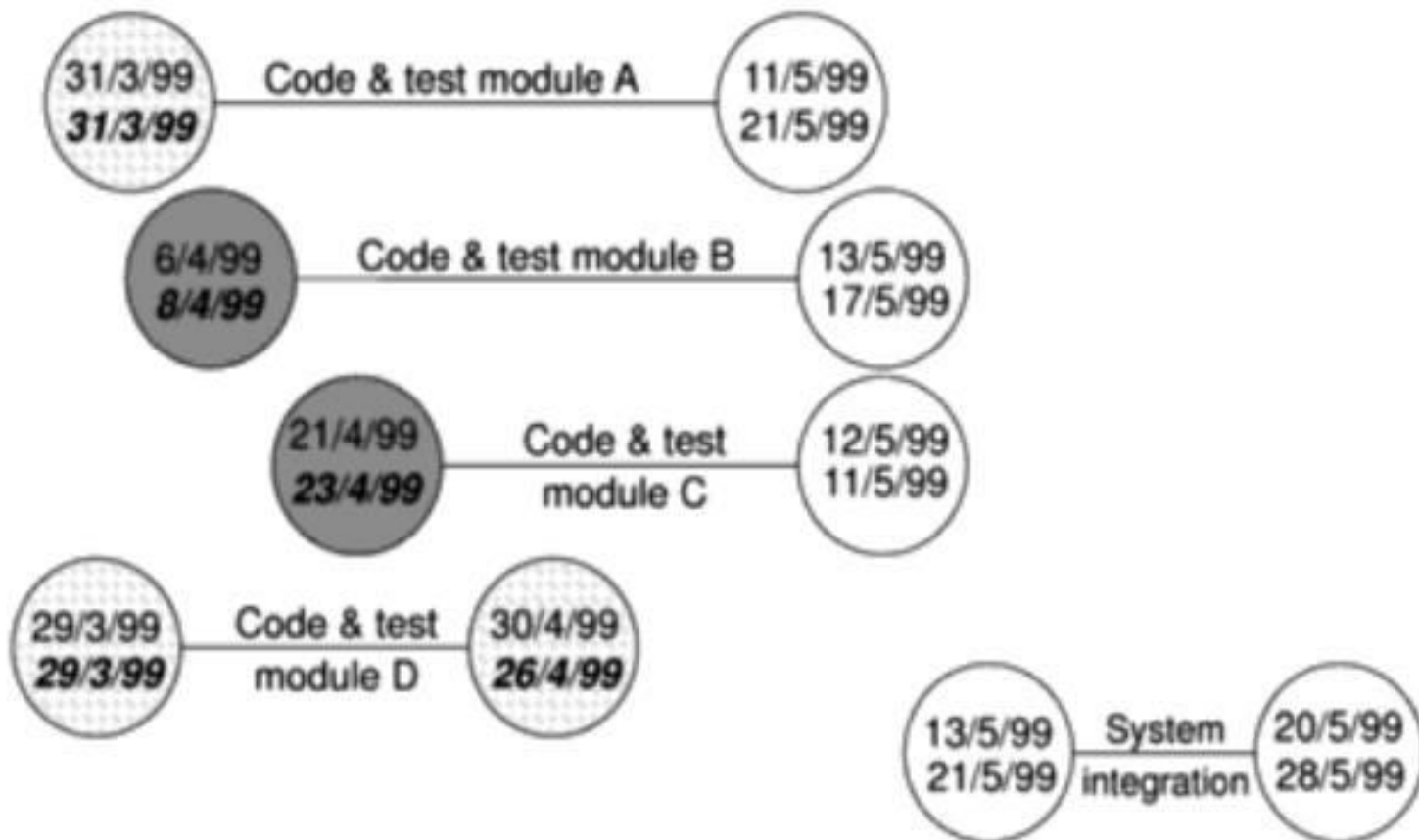
Slip Chart_Example



Ball Chart

- A more prominent way of showing whether or not targets have been met is to use a ball chart.
- In this technique of visualization of progress, the circles indicate start and completion points for activities.
- The circles initially contain the original scheduled dates.
- Whenever revisions are produced, these are added as second dates in the appropriate circle when:
 - *an activity is actually started or completed*
 - *the relevant date replaces the revised estimate.*

Ball Chart_Example

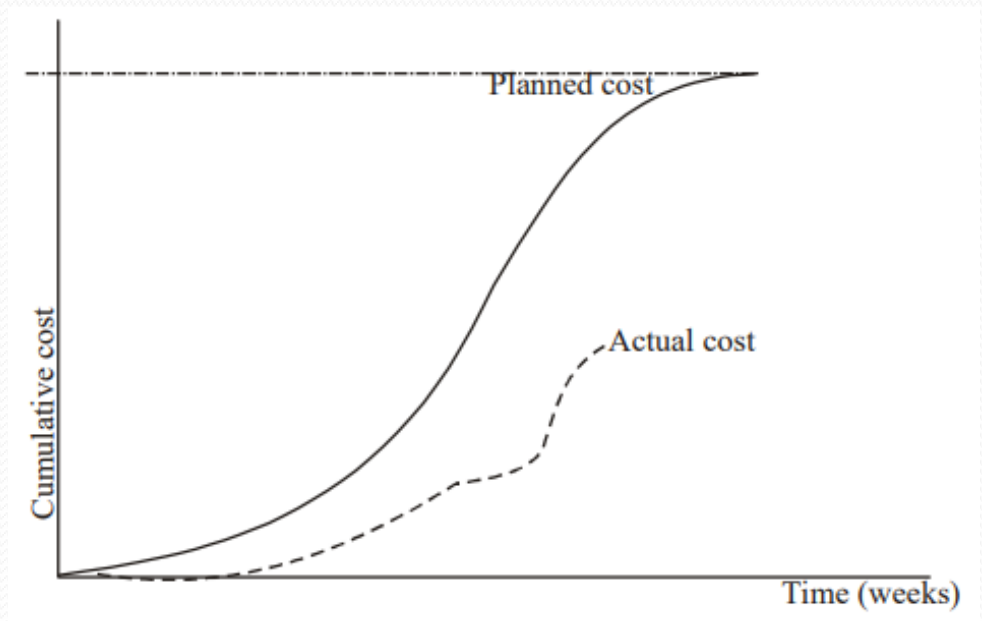


Cost Monitoring

- Expenditure monitoring is a vital component of project control because it provides an indication of the effort that has gone into a project.
- A project might be on time but only because more money has been spent on activities than originally budgeted
- Expenditure (Cost) monitoring is an important component of project control, not only in itself, but also because it provides an indication of the effort that has gone into (or at least been charged to) a project.

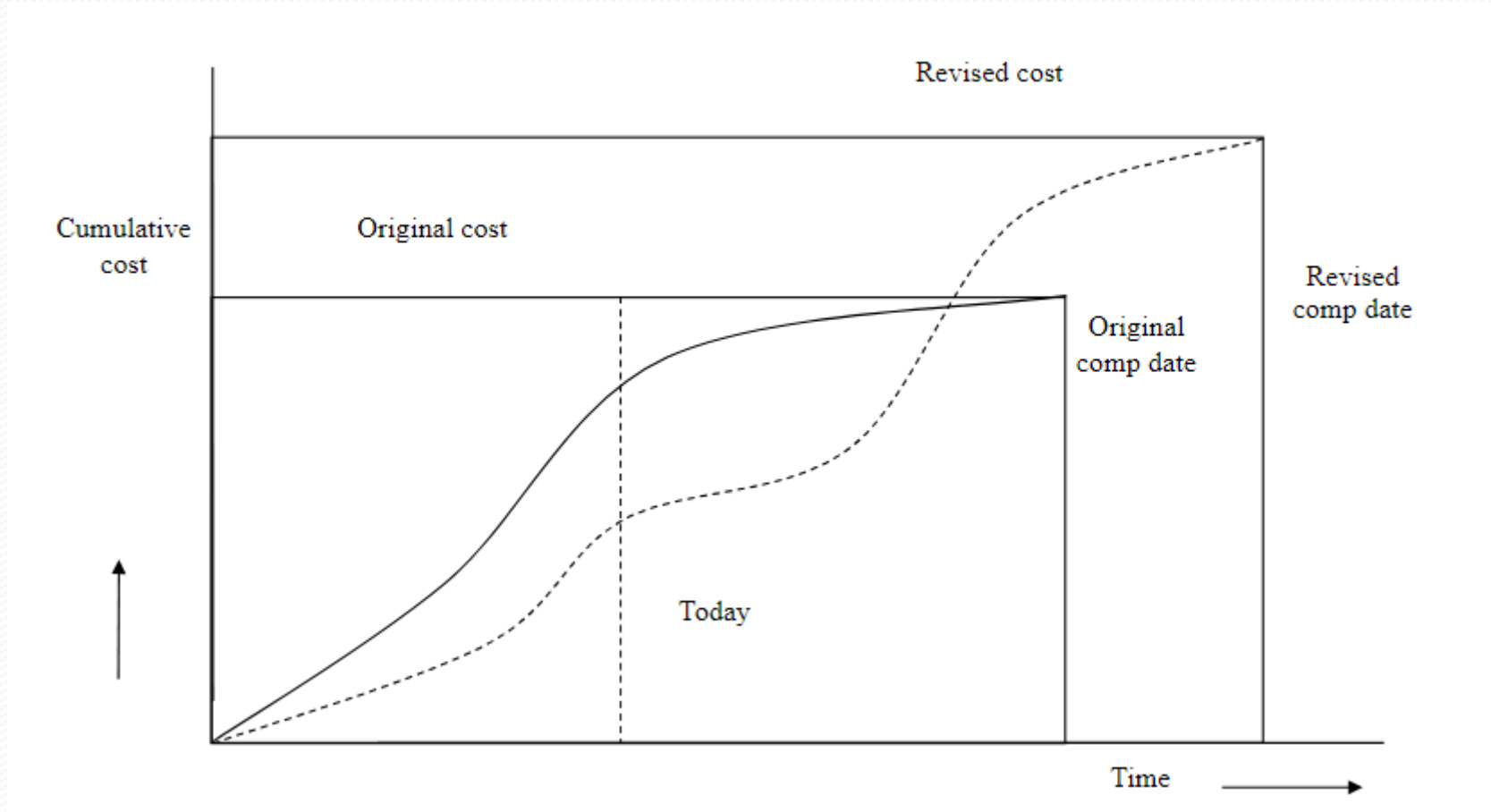
Cost Monitoring

- A cumulative expenditure chart provides a simple method of comparing actual and planned expenditure.
- Figure below illustrates a project that is running late or one that is on time but has shown substantial costs savings.



Tracking Cumulative Expenditure

Cost Monitoring



Cost graph with cost /time Extension

Earned Value Analysis (EVA)

- Earned Value Analysis (EVA) is a cost monitoring technique which integrates cost, schedule and scope and can be used to forecast future performance and project completion dates.
- EVA is based on assigning a 'value' to each task or work package as identified in the work breakdown structure (WBS) based on the original expenditure forecasts.
- A task that has not started is assigned the value zero and when it has been completed, it is credited with the value of the task.

Earned Value Analysis (EVA)

- Three quantities form the basis for cost performance measurement using Earned Value Analysis.
- They are:
 - *Budgeted Cost of Work Scheduled (BCWS) or Planned Value (PV)*
 - *Budgeted Cost of Work Performed (BCWP) or Earned Value (EV) and*
 - *Actual Cost of Work Performed (ACWP) or Actual Cost (AC).*

Earned Value Analysis (EVA)

- *Budgeted Cost of Work Scheduled (BCWS) or Planned Value (PV):*
 - The sum of budgets for all work packages scheduled to be accomplished within a given time period.
- *Budgeted Cost of Work Performed (BCWP) or Earned Value (EV):*
 - The sum of budgets for completed work packages and completed portions of open work packages.
- *Actual Cost of Work Performed (ACWP) or Actual Cost (AC):*
 - The actual cost incurred in accomplishing the work performed within a given time period.

Earned Value Analysis (EVA)

- Additional terms are defined to record cost and schedule performance and program budget:

- ***Schedule Variance (SV)***

- The difference between the work actually performed (BCWP) and the work scheduled (BCWS).

$$SV = BCWP - BCWS$$

- ***Cost Variance (CV)***

- The difference between the planned cost of work performed (BCWP) and actual cost incurred for the work (ACWP).

$$CV = BCWP - ACWP$$

Earned Value Analysis (EVA)

- Two Performance Ratios are calculated as:
 - ***Cost Performance Index (CPI)***
 - The ratio of cost of work performed (BCWP) to actual cost (ACWP).
 - CPI of 1.0 implies that the actual cost matches to the estimated cost.
 - CPI greater than 1.0 indicates work is accomplished for less cost than what was planned or budgeted.
 - CPI less than 1.0 indicates the project is facing cost overrun.

$$CPI = \frac{BCWP \text{ or } EV}{ACWP \text{ or } AC}$$

Earned Value Analysis (EVA)

- ***Schedule Performance Index (SPI)***
 - The ratio of work accomplished (BCWP) versus work planned (BCWS), for a specific time period.
 - SPI indicates the rate at which the project is progressing.

$$SPI = \frac{BCWP \text{ or } EV}{BCWS \text{ or } PV}$$

***Example_1:** Perform Earned Value Analysis of the given project*

Activity	Duration (Days)	Precedence	Cost/Day
A	3	-	500
B	3	A	100
C	4	B	400
D	2	B	500
E	3	D	500

The progress after the 8th day is:

Activity	% Completion	Incurred Cost
A	100	2000
B	100	500
C	25	500
D	50	500
E	0	0

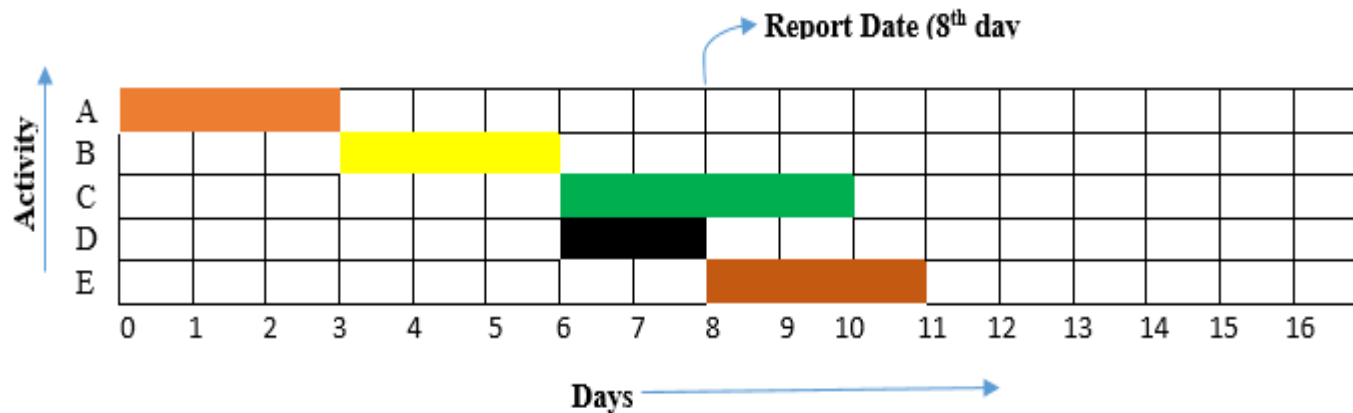
Calculate SV, CV, SPI and CPI

Solution

Activity	Duration (Days)	Precedence	Cost/Day	Total Cost
A	3	-	500	1500
B	3	A	100	300
C	4	B	400	1600
D	2	B	500	1000
E	3	D	500	1500
Total	13			

* $Total\ Cost = Duration * Cost/Day$

The Gantt Chart is as follows:



Now

Activity	ACWP	BCWP	BCWS
A	2000	1500	1500
B	500	300	300
C	500	400	800
D	500	500	1000
E	0	0	0
Total	3500	2700	3600

* $BCWP = \text{Total Cost} * \% \text{ Completion}$

- *BCWS is calculated according to the activity completed within reporting days i.e. 8 days.*
- *For activity A and B, these both are completed until 8th day so BCWS is same as BCWP*
- *For activity C, the task is complete only for 2 days from gantt chart and 1 day cost is 400 as total cost of C is 1600 and total duration is 4 days.*
- *For activity D, the task is for 2 days from gantt and 1 day cost is 500 as total cost of D is 1000 and total duration is 2 days.*

Now

$$\begin{aligned}SV &= BCWP - BCWS \\&= 2700 - 3600 \\&= -900\end{aligned}$$

$$\begin{aligned}CV &= BCWP - ACWP \\&= 2700 - 3500 \\&= -800\end{aligned}$$

$$SPI = \frac{BCWP}{BCWS} = \frac{2700}{3600} = 0.75$$

$$CPI = \frac{BCWP}{ACWP} = \frac{2700}{3500} = 0.771$$

Example_2: Perform Earned Value Analysis of the given project

Activity	Duration (Days)	Precedence	Cost/Day
A	8	-	200
B	4	-	200
C	4	A,B	300
D	4	B	400

The progress after the 12th day is:

Activity	% Completion	Incurred Cost
A	100	1400
B	100	400
C	75	1000
D	50	1000

Calculate SV, CV, SPI and CPI

Answer: Example_2

$$\begin{aligned}SV &= BCWP - BCWS \\&= 4100 - 5200 \\&= -1100\end{aligned}$$

$$\begin{aligned}CV &= BCWP - ACWP \\&= 4100 - 3800 \\&= 300\end{aligned}$$

$$SPI = \frac{BCWP}{BCWS} = \frac{4100}{5200} = 0.788$$

$$CPI = \frac{BCWP}{ACWP} = \frac{4100}{3800} = 1.078$$

Example_3: Perform Earned Value Analysis of the given project

Activity	Duration (Days)	Precedence	Cost/Day
A	5	-	200
B	5	A	200
C	10	A	100
D	5	A, B	200

The progress after the 10th day is:

Activity	% Completion	Incurred Cost
A	100	1200
B	100	800
C	40	800
D	0	0

Calculate SV, CV, SPI and CPI

Answer: Example_3

$$\begin{aligned}SV &= BCWP - BCWS \\&= 2400 - 2500 \\&= -100\end{aligned}$$

$$\begin{aligned}CV &= BCWP - ACWP \\&= 2400 - 2800 \\&= -400\end{aligned}$$

$$SPI = \frac{BCWP}{BCWS} = \frac{2400}{2500} = 0.96$$

$$CPI = \frac{BCWP}{ACWP} = \frac{2400}{2800} = 0.85$$

Project Controlling

- Project controlling is the second stage of this project management phase.
- It's the process of taking any action needed for issues or changes that have been identified during the monitoring stage.
- This is about putting controls in place to ensure that things don't go further off track, but also taking action anything that needs fixing.
- This is compared to project monitoring which is focused on observation and evaluation.

Project Controlling

- Project controlling activities include:
 - **Analyzing the data and information** that you've collected during the project monitoring stage
 - **Assessing any changes from your original plan**, and the impact of these variances on the project's course to success
 - **Prioritizing activities** according to their potential impact on your project
 - **Making decisions about the best course of action** to get the project back on track. This includes controlling decisions about:
 - Any changes to your plan
 - Any changes to your overall costs or budget
 - Any changes to the schedule or timeline of the project

Project Controlling

- **Delivering any updated documentation, such as revised project schedules**
- **Informing and negotiating with key stakeholders as needed throughout the project**
- *Simple difference among monitoring and controlling is:*
 - **Monitoring is focused on observing and tracking the project's progress**
 - This is to ensure the project is on track and tasks are being completed on time.
 - **Controlling is focused on taking any corrective measures necessary.**
 - This is to ensure that the project meets its goals and objectives.



End of Chapter 6

Thank You !!!!